Better approximation to the susie-love model

1 Model discussion

Last time we considered the susie-relax model, where the uncertainty of R only affects the mean, not the variance:

$$\begin{cases} x|R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, \bar{R}/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \tag{1}$$

which leads to the conditional distribution:

$$p(x_{-j} \mid x_j, \beta, \gamma = 1) = \mathcal{N}\left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{\beta^2}{\nu_0 + 3}\right) \bar{S}_j\right), \tag{2}$$

where $\bar{S}_j$ is the Schur complement of $\bar{R}_{jj}$ in $\bar{R}$. Because this depends on β, this model does not satisfy the "irrelevance of null SNPs" property. Is there another model that arises from the conditional distribution:

$$p(x_{-j} \mid x_j, \beta, \gamma = j) = \mathcal{N}\left(x_j \bar{R}_{j,-j}, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0 + 3}\right) \bar{S}_j\right)? \tag{3}$$

2 Some new models with better approximation to susie-love

Recall our original SER model from 'susie-love':

$$\begin{cases} x|R, \beta, \gamma \sim \mathcal{N}(\beta R_\gamma, R/N) \\ R \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1). \end{cases} \tag{4}$$

In the following, we show that it satisfies "irrelevance of the null SNPs" and, in fact, leads to conditional distribution similar to (3). The novolty here is what we learned from susie-relax last week: To represent R in terms of blocks.

2.1 Model A (equivalent to the original SER-love model)

Let's write down $p(x|R, \beta, \gamma = 1)$, other cases of γ are similar. Let

$$d_1 = R_{11} \in \mathbb{R}_+, \quad z_1 = \frac{R_{-1,1}}{d_1} \in \mathbb{R}^{J-1},$$

and

$$S_1 = R_{-1,-1} - d_1 z_1 z_1^\top$$

be the Schur's complement of R_{11} in R. Hence, the latent R is represented as

$$R = \begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}. \tag{5}$$

Similarly, denote

$$\bar{S}_1 = \bar{R}_{-1,-1} - \bar{R}_{-1,1}\bar{R}_{-1,1}^\top$$

be the Schur's complement of $\bar{R}_{11}$ in $\bar{R}$, and $\mu_1 = \bar{R}_{-1,1}$, we can write

$$\bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}. \tag{6}$$

Properties of IW distribution (together with the fact that $\bar{R}_{11} = 1$) imply:

$$\begin{cases} d_1 \sim \mathrm{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\bar{R}_{-1,1}, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \end{cases} \tag{7}$$

The original SER version of Susie-love (with $\gamma = 1$; similar for other γ) becomes:

$$\begin{cases} x | \beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta \begin{pmatrix} d_1 \\ d_1 z_1 \end{pmatrix}, \frac{1}{N}\begin{pmatrix} d_1 & d_1 z_1^\top \\ d_1 z_1 & S_1 + d_1 z_1 z_1^\top \end{pmatrix}\right) \\ d_1 \sim \mathrm{IG}\left(\frac{\nu_0 + 2}{2}, \frac{\nu_0}{2}\right) \\ z_1 | S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0 \bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \tag{8}$$

Write $x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}$. From the block normal distribution in (8), we have

$$p(x \mid \gamma = 1, \beta, d_1, z_1, S_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} \mid x_1 z_1, \frac{S_1}{N}\right).$$

The key simplification is that, **after conditioning on x_1, the conditional mean of x_{-1} is $x_1 z_1$ and no longer contains β or d_1** (relevant to our earlier discussion). Now we proceed to show that this satisfies the "irrelevance of null SNPs" property:

1. Marginalize out $z_1 | S_1$ leads to:

$$p(x \mid \gamma = 1, \beta, d_1, S_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} \mid x_1 \mu_1, \left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right) S_1\right). \tag{9}$$

What I found so interesting is that the conditional distribution $p(x_{-1} \mid x_1, S_1, \gamma = 1)$ resembles what was suggested by Matthew last time (cf. (3)).

2. Marginalize over S_1. Thus

$$p(x \mid \gamma = 1, \beta, d_1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{d_1}{N}\right) t_{J-1,\nu_0+3}\left(x_{-1} \mid x_1\mu_1, \frac{\nu_0}{\nu_0+3}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)\bar{S}_1\right), \tag{10}$$

where the second argument is the multivariate t scale matrix and is denoted by $T_1(x)$:

$$T_1(x) \propto |\bar{S}_1|^{-1/2}\left(\frac{1}{N} + \frac{x_1^2}{\nu_0}\right)^{-(J-1)/2}\left[1 + \frac{(x_{-1} - x_1\mu_1)^\top(\nu_0\bar{S}_1)^{-1}(x_{-1} - x_1\mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}}\right]^{-(\nu_0+J+2)/2}.$$

Using the block inverse formula (notice the block form of $\bar{R}$ (6)),

$$x^\top \bar{R}^{-1} x = x_1^2 + (x_{-1} - x_1\mu_1)^\top \bar{S}_1^{-1}(x_{-1} - x_1\mu_1),$$

so

$$1 + \frac{(x_{-1} - x_1\mu_1)^\top(\nu_0\bar{S}_1)^{-1}(x_{-1} - x_1\mu_1)}{\frac{1}{N} + \frac{x_1^2}{\nu_0}} = \frac{x^\top\bar{R}^{-1}x + \nu_0/N}{x_1^2 + \nu_0/N}.$$

Therefore

$$T_1(x) \propto |\bar{S}_1|^{-1/2}\left(x_1^2 + \frac{\nu_0}{N}\right)^{(\nu_0+3)/2}\left(x^\top\bar{R}^{-1}x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2}. \tag{11}$$

When comparing different values of γ, **the final factor is common across SNPs.** Also, since $\bar{R}_{11} = 1$, $|\bar{R}| = |\bar{S}_1|$; analogously, $|\bar{S}_j| = |\bar{R}|$ for any SNP j with $\bar{R}_{jj} = 1$.

3. Marginalize over β and d_1 finally leads to

$$\boxed{p(x \mid \gamma = 1) = T_1(x)\int_0^\infty \mathcal{N}\left(x_1 \mid 0, \frac{d_1}{N} + d_1^2\sigma_0^2\right) d\,\mathrm{IG}\left(d_1; \frac{\nu_0+2}{2}, \frac{\nu_0}{2}\right)}.$$

This is a one-dimensional integral over $d_1 > 0$ which does not have closed-form expression. Cancel out the first (determinant term) and the last (quadratic term) in $T_1(x)$ because they are common for all SNPs, we have

$$\boxed{p(\gamma = j \mid x) \propto \left(1 + \frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2}\int_0^\infty \mathcal{N}\left(x_j \mid 0, \frac{d_j}{N} + d_j^2\sigma_0^2\right) d\,\mathrm{IG}(d_j)}. \tag{12}$$

Conclusion. The resulting PIP weight depends on SNP j only through x_j.

2.2 Model B

Simplify $d_1 \equiv 1$ to avoid intractable integral (and we know the diagonal of the in-sample LD matrix is 1 anyway), we have the following model (model B):

$$\begin{cases} x|\beta, \gamma = 1, z_1, S_1 \sim \mathcal{N}\left(\beta\begin{pmatrix}1\\ z_1\end{pmatrix}, \frac{1}{N}\begin{pmatrix}1 & z_1^\top \\ z_1 & S_1 + z_1 z_1^\top\end{pmatrix}\right) \\ z_1|S_1 \sim \mathcal{N}\left(\mu_1, \frac{S_1}{\nu_0}\right) \\ S_1 \sim \mathcal{IW}(\nu_0\bar{S}_1, \nu_0 + J + 1) \\ \beta \sim \mathcal{N}(0, \sigma_0^2) \end{cases} \tag{13}$$

This model leads to

$$\boxed{p(x \mid \gamma=1) \propto |\bar{R}|^{-1/2} \left(1+\frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \left(x^\top \bar{R}^{-1} x + \frac{\nu_0}{N}\right)^{-(\nu_0+J+2)/2} \mathcal{N}\left(x_1 \mid 0, \frac{1}{N}+\sigma_0^2\right)}.$$

Since $|\bar{R}|$ and $x^\top \bar{R}^{-1} x$ do not depend on which SNP is chosen, the PIP weights simplify further. Dropping the common factors gives

$$\boxed{p(\gamma=j \mid x) \propto \left(1+\frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} \mathcal{N}\left(x_j \mid 0, \frac{1}{N}+\sigma_0^2\right)}. \tag{14}$$

Compare to SER-RSS's PIP. Recall SER-RSS's PIP:

$$\tilde{p}(\gamma=j \mid x) \propto \exp\left(\frac{N\sigma_0^2}{N\sigma_0^2+1}\frac{Nx_j^2}{2}\right) \tag{15}$$

We now show that (14) tends to (15) as $\nu_0 \to \infty$. Since

$$\left(1+\frac{Nx_j^2}{\nu_0}\right)^{(\nu_0+3)/2} = \left(1+\frac{2}{\nu_0+3} \times \frac{\nu_0+3}{\nu_0}\frac{Nx_j^2}{2}\right)^{(\nu_0+3)/2} \asymp \exp\left\{\frac{\nu_0+3}{\nu_0}\frac{Nx_j^2}{2}\right\}$$

Also,

$$\mathcal{N}\left(x_j \mid 0, \frac{1}{N}+\sigma_0^2\right) \propto \exp\left\{-\frac{x_j^2}{2(1/N+\sigma_0^2)}\right\} = \exp\left\{-\frac{1}{1+N\sigma_0^2}\frac{Nx_j^2}{2}\right\}.$$

Therefore the limiting weight in (14) is asymptotically (as ν_0 gets large)

$$\exp\left\{\left(\frac{\nu_0+3}{\nu_0} - \frac{1}{N\sigma_0^2+1}\right)\frac{Nx_j^2}{2}\right\} = \exp\left\{\left(\frac{N\sigma_0^2}{N\sigma_0^2+1} - \frac{1}{\nu_0+3}\right)\frac{Nx_j^2}{2}\right\} \approx \exp\left\{\frac{N\sigma_0^2}{N\sigma_0^2+1}\frac{Nx_j^2}{2}\right\}$$

which is exactly the SER-RSS PIP weight in (15). Hence, for finite ν_0, PIP under SER-love model (B) is slightly more diffused than susie-rss.

Inference under model B (13). For a general SNP j, write

$$\mu_j = \bar{R}_{-j,j}, \qquad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \qquad \kappa_j = \nu_0 + Nx_j^2.$$

Under model B, conditional on $\gamma = j$, the likelihood factorizes as

$$p(x \mid \beta, z_j, S_j, \gamma=j) = \mathcal{N}\left(x_j \mid \beta, \frac{1}{N}\right) \mathcal{N}_{J-1}\left(x_{-j} \mid x_j z_j, \frac{S_j}{N}\right).$$

Thus β is conditionally independent of (z_j, S_j) given x and $\gamma = j$, and

$$p(\beta, z_j, S_j \mid x, \gamma=j) = p(\beta \mid x_j, \gamma=j)\, p(z_j, S_j \mid x, \gamma=j).$$

- The posterior for β is the usual normal-normal update:

$$\boxed{\beta \mid x, \gamma = j \sim \mathcal{N}\left(\frac{N}{N + \sigma_0^{-2}} x_j, \ \frac{1}{N + \sigma_0^{-2}}\right).}$$

- For the LD-column variables, first condition on S_j. Combining

$$z_j \mid S_j \sim \mathcal{N}_{J-1}\left(\mu_j, \frac{S_j}{\nu_0}\right), \qquad x_{-j} \mid x_j, z_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j z_j, \frac{S_j}{N}\right)$$

gives

$$\boxed{z_j \mid S_j, x, \gamma = j \sim \mathcal{N}_{J-1}\left(\frac{\nu_0 \mu_j + N x_j x_{-j}}{\nu_0 + N x_j^2}, \ \frac{S_j}{\nu_0 + N x_j^2}\right).}$$

- Integrating out z_j gives

$$x_{-j} \mid x_j, S_j, \gamma = j \sim \mathcal{N}_{J-1}\left(x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) S_j\right).$$

Therefore the posterior for S_j is inverse-Wishart:

$$\boxed{S_j \mid x, \gamma = j \sim \mathcal{IW}\left(\nu_0 \bar{S}_j + \frac{N\nu_0}{\nu_0 + N x_j^2}(x_{-j} - x_j \mu_j)(x_{-j} - x_j \mu_j)^\top, \ \nu_0 + J + 2\right).}$$

Equivalently, the joint posterior of (z_j, S_j) can be represented by first drawing S_j from this inverse-Wishart distribution and then drawing z_j from the conditional normal distribution above. Finally, the full posterior over γ uses the PIP weights in (14), normalized over $j = 1, \ldots, J$. The prior parameter σ_0^2 and ν_0 can also be inferred from E-M algorithm given these closed-form posteriors.

2.3 Discussion

A caveat to our previously discussed conditional distribution. Note that in equation (9), we have

$$\begin{cases} x_{-j} | x_j, \gamma = j, S_j \sim \mathcal{N}_{J-1}\left(x_{-j} \mid x_j \mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) S_j\right) \\ S_j \sim \mathcal{IW}(\nu_0 \bar{S}_j, \nu_0 + J + 1) \end{cases} \tag{16}$$

which leads to

$$p(x_{-j} \mid x_j, \gamma = j) = t_{J-1, \nu_0 + 3}\left(x_{-j} \mid x_j \mu_j, \frac{\nu_0}{\nu_0 + 3}\left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right) \bar{S}_j\right) \tag{17}$$

This t-distribution has a nice property that it factors in to two parts, one only depends on x_j and the other only depends on $x^\top (\bar{R})^{-1} x$ and $|\bar{R}|$ which can be cancelled out when comparing evidence between different j (equation (11)). But if we directly use

$$
\begin{aligned}
p(x_{-j} \mid x_j, \gamma = j) &= \mathcal{N}_{J-1}\left(x_j\mu_j, \left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right)\bar{S}_j\right) \\
&\propto |\bar{R}|^{-1/2}\left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right)^{-(J-1)/2} \exp\left\{-\frac{1}{2}\frac{(x_{-j} - x_j\mu_j)^\top(\nu_0\bar{S}_j)^{-1}(x_{-j} - x_j\mu_j)}{\frac{1}{N} + \frac{x_j^2}{\nu_0}}\right\} \\
&= |\bar{R}|^{-1/2}\left(\frac{1}{N} + \frac{x_j^2}{\nu_0}\right)^{-(J-1)/2} \exp\left\{-\frac{1}{2}\frac{x^\top(\bar{R})^{-1}x - x_j^2}{\frac{1}{N} + \frac{x_1^2}{\nu_0}}\right\}
\end{aligned}
$$

then the terms $x^\top(\bar{R})^{-1}x$ and x_j are still coupled, which makes $x^\top(\bar{R})^{-1}x$ fails to get cancelled out. However, this Normal distribution approximates (17) well (we can show this using the approximation $(1 + y/\nu)^\nu \approx e^y$ for moderate to large ν)

Next? I think there are two routes:

1. Extending model B to a susie-like model, which seems pretty hard because the L SER model might not have the same covariance (so we may not easily have IBSS-like algorithm).
2. Go with the susie-relax model last week and show that its PIP approximates the PIP in SER-love model well, therefore having "approximated" irrelevance of null SNPs property.